Omar Ramadan

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EDUCATION

Bachelor of Software Engineering London, ON
Western University *Sep. 2022 – May 2026*
Relevant Coursework: Project Management, Networking, Database Management, Requirements Analysis, Software Design

EXPERIENCE

Computer Vision Jr. Lead September 2024 – Present
Western Aerospace Design Team *London, ON*

- Designed a computer vision system using **Python** and **OpenCV** for object detection and landing zone identification on a tricopter, resulting in improved **accuracy by 30%** during autonomous flight testing.
- Developed offboard flight task modules using **PX4** and **MAVLink** protocols to relay vision detection data, offloading tasks to minimize failures and maintain mission-critical integrity.
- Conducted a hands-on demo on prototype **detection** software, **engaging** team members through code explanation, resulting in a **20%** improvement in task **completion** rates and prolonged club attendance.

Software Engineer – DataCareLLC September – August 2025
RBC Initiative Project | RIPPEN *Remote Work*

- Built an offline Java application with H2 database for mobile medical teams, enabling 200+ healthcare workers to register and treat thousands of patients in remote Indian regions lacking electricity or internet.
- Engineered a queue-based sync system from local H2 database nodes to a cloud-hosted YugabyteDB cluster, scaling across distributed networks; deployed backend infrastructure on GCP for regional clinic access and data sharing.
- Developed a touchscreen-friendly frontend using Vite.js and coordinated remotely with the client to align UX with field use, ensuring usability on low-power devices in resource-limited settings.

PROJECTS

DeepEnd | *Next.js, OpenAI, Firebase, Vercel, TensorFlow*

- Created a platform with tailored project recommendations for aspiring software engineers using **Next.js**, deployed on **Vercel**, attracting **200+ users** and reducing tech career entry barriers by offering skill-matched projects**.**
- Designed a secure authentication system using **Firebase Auth** and **Firestore**, enabling flexible login options via **Google** and **GitHub**, with **95%** user adoption.
- Engineered personalized project recommendations by integrating a **Siamese TensorFlow** model for skill classification and **OpenAI API** for tailored content, enhancing user satisfaction by **30%** through skill-aligned suggestions.
- Secured **1st** place out of 30+ teams in the GDSC Hackathon for **Best Overall Hackathon**, demonstrating innovative use of **TensorFlow** and OpenAI in project alignment.

Folium | *Svelte, MySQL, C++, PostgreSQL, Vercel, AWS, Multithreading*

- Built a **collaborative** note-taking platform for enrolled students to maintain a single unified note. Using **Svelte** and a multithreaded **C++** server with PostgreSQL, enabling real-time shared editing and concurrency-safe note processing through **ACID** transactions and **mutex-based** synchronization.
- Engineered a scalable access control system using a **dispatcher-thread pool model**, including **DAL**, to securely manage user sessions via JWTs—enabling efficient load distribution across a multi-user environment.
- Deployed a distributed full-stack system with a globally hosted Svelte frontend (Vercel) and a C++ backend on **AWS**, enabling public access and supporting over **50 concurrent users** with stable performance under simulated load conditions.

Western Engineering Competition | *Arduino, ESP8226 (WiFi board), IP Hosting, C++*

- **Led** a team of 4 to design a **remote-controlled vehicle** for navigating disaster-affected terrains, **meeting** competition criteria for **speed** and **efficiency**.
- Engineered a **servo-driven claw** with bi-directional control, enabling precision **handling of 1 kg objects** and enhancing adaptability in disaster scenarios.
- Controlled the vehicle via a **WiFi board** with a hosted client, extending **range by 20%** over traditional radio methods
- **Secured** a **top 10** placement out of **50+ teams** participating teams, placing in the **top 10** for innovative design.

TECHNICAL SKILLS

Languages: Java, C++, C#, Typescript, SQL, Python, HTML/CSS, Arduino
Frameworks: Node.js, Next.js, Apache Derby, React, TensorFlow, Vite.js
Developer Tools: Git, VS Code, Visual Studio, IntelliJ, Unity, Vim, Ardunio IDE, AWS Cloud, Cube-IDE, GCP,
Libraries: NumPy, Matplotlib, Flask, Maven, TailwindCSS, OpenCV, python-can